

Venus Labs - EBrake & DeviationSentinel Changes

Security Assessment

CertiK Assessed on Apr 9th, 2026





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The security assessment was prepared by Certik.

Executive Summary

TYPES

Lending

ECOSYSTEM

Binance Smart Chain
(BSC)

METHODS

Manual Review, Static Analysis

LANGUAGE

Solidity

TIMELINE

Preliminary comments published on 04/08/2026

Final report published on 04/09/2026

Vulnerability Summary



2

Total Findings

2

Resolved

0

Partially Resolved

0

Acknowledged

0

Declined



0 Centralization

Centralization findings highlight privileged roles & functions and their capabilities, or instances where the project takes custody of users' assets.



0 Critical

Critical risks are those that impact the safe functioning of a platform and must be addressed before launch. Users should not invest in any project with outstanding critical risks.



0 Major

Major risks may include logical errors that, under specific circumstances, could result in fund losses or loss of project control.



0 Medium

Medium risks may not pose a direct risk to users' funds, but they can affect the overall functioning of a platform.



2 Minor

2 Resolved



Minor risks can be any of the above, but on a smaller scale. They generally do not compromise the overall integrity of the project, but they may be less efficient than other solutions.



0 Informational

Informational errors are often recommendations to improve the style of the code or certain operations to fall within industry best practices. They usually do not affect the overall functioning of the code.

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Disclaimer

CODEBASE | VENUS LABS - EBRAKE & DEVIATIONSENTINEL CHANGES

Repository

The changes introduced in <https://github.com/VenusProtocol/venus-periphery/pull/60> and <https://github.com/VenusProtocol/venus-periphery/pull/62/changes/7fab69f5a583577331230e646abebfb4e2b5a41> were audited.

Commit

[40117018a5cec078429ca10bc1163ee5e8d608dd](#)

AUDIT SCOPE**VENUS LABS - EBRAKE & DEVIATIONSENTINEL
CHANGES**

VenusProtocol/venus-periphery



DeviationSentinel/DeviationSentinel.sol



EmergencyBrake/EBrake.sol



EmergencyBrake/IEBrake.sol

APPROACH & METHODS

VENUS LABS - EBRAKE & DEVIATIONSENTINEL CHANGES

This audit was conducted for Venus Labs to evaluate the security and correctness of the smart contracts associated with the Venus Labs - EBrake & DeviationSentinel Changes project. The assessment included a comprehensive review of the in-scope smart contracts. The audit was performed using a combination of Manual Review and Static Analysis.

The review process emphasized the following areas:

- Architecture review and threat modeling to understand systemic risks and identify design-level flaws.
- Identification of vulnerabilities through both common and edge-case attack vectors.
- Manual verification of contract logic to ensure alignment with intended design and business requirements.
- Dynamic testing to validate runtime behavior and assess execution risks.
- Assessment of code quality and maintainability, including adherence to current best practices and industry standards.

The audit resulted in findings categorized across multiple severity levels, from informational to critical. To enhance the project's security and long-term robustness, we recommend addressing the identified issues and considering the following general improvements:

- Improve code readability and maintainability by adopting a clean architectural pattern and modular design.
- Strengthen testing coverage, including unit and integration tests for key functionalities and edge cases.
- Maintain meaningful inline comments and documentations.
- Implement clear and transparent documentation for privileged roles and sensitive protocol operations.
- Regularly review and simulate contract behavior against newly emerging attack vectors.

FINDINGS

VENUS LABS - EBRAKE & DEVIATIONSENTINEL
CHANGES

2

Total Findings

0

Critical

0

Centralization

0

Major

0

Medium

2

Minor

0

Informational

This report has been prepared for Venus Labs to identify potential vulnerabilities and security issues within the reviewed codebase. During the course of the audit, a total of 2 issues were identified. Leveraging a combination of Manual Review & Static Analysis the following findings were uncovered:

ID	Title	Category	Severity	Status
VLE-01	Oracle Reverts Can Denial-Of-Service <code>handleDeviation()</code> Emergency Actions	Logical Issue	Minor	● Resolved
VLE-02	<code>market.underlying()</code> Call Assumes ERC20 Interface, Breaking Native Asset Markets	Logical Issue	Minor	● Resolved

VLE-01 | Oracle Reverts Can Denial-Of-Service `handleDeviation()` Emergency Actions

Category	Severity	Location	Status
Logical Issue	● Minor	DeviationSentinel/DeviationSentinel.sol (base): 219–221	● Resolved

Description

The `handleDeviation()` function relies on direct calls to `RESILIENT_ORACLE.getPrice()` and `SENTINEL_ORACLE.getPrice()` without error handling. If either oracle reverts (due to misconfiguration, network issues, or upstream failures), the entire function will revert, preventing emergency protective actions from being executed.

This creates a critical failure mode where the sentinel becomes non-functional precisely when it's needed most—during oracle instability or price manipulation events. The contract's core safety mechanism can be completely bypassed by causing oracle calls to revert.

```
function checkPriceDeviation(...) public view returns (...) {
    // ...
    oraclePrice = RESILIENT_ORACLE.getPrice(underlyingToken);
    sentinelPrice = SENTINEL_ORACLE.getPrice(underlyingToken);
    // ...
}
```

Scenario

1. Configure a token with deviation monitoring enabled
2. Cause either the ResilientOracle or SentinelOracle to revert when queried (e.g., through misconfiguration or upstream dependency failure)
3. Keeper calls `handleDeviation()` for the affected market
4. The function reverts at the oracle call, preventing any emergency actions from being taken

Recommendation

Wrap oracle price calls in try/catch blocks. Treat oracle failures as deviations to trigger conservative protective actions, or implement a fallback mechanism that pauses affected markets when oracle availability cannot be verified.

Alleviation

[Venus Labs, 04/09/2026]:

This is by design. The sentinel's purpose is to detect a discrepancy between two oracles, if either reverts, there's no discrepancy to measure, so taking no action is the correct behaviour. Wrapping in try/catch and acting on a revert would

mean the sentinel fires based on oracle unavailability, not price anomaly a different signal entirely and not a safe trigger for pausing markets. The ResilientOracle already handles upstream failures internally by aggregating multiple sources; it only reverts when it genuinely cannot produce a reliable price. In that case, emergency response belongs to the multisig, not an automated sentinel acting on incomplete data.

VLE-02 `market.underlying()` Call Assumes ERC20 Interface, Breaking Native Asset Markets

Category	Severity	Location	Status
Logical Issue	Minor	DeviationSentinel/DeviationSentinel.sol (base): 185~186	Resolved

Description

The contract assumes all `IVToken` markets implement an `underlying()` function that returns an ERC20 token address. This assumption fails for native-asset markets (`vBNB`) where the underlying is the chain's native currency and the `underlying()` function may revert.

In `handleDeviation()` and `checkPriceDeviation()`, the code calls `market.underlying()` and uses the result as a key in `tokenConfigs` mapping:

```
address underlyingToken = market.underlying();
DeviationConfig memory config = tokenConfigs[underlyingToken];
```

If `underlying()` reverts for native-asset markets, the entire deviation check and emergency action mechanism will fail for those markets. Even if it returns `address(0)`, the `tokenConfigs[address(0)]` lookup will can't be configured.

Scenario

1. Deploy DeviationSentinel with vBNB market configured
2. Keeper calls `handleDeviation(vBNB)`
3. `vBNB.underlying()` reverts
4. Function fails, preventing price deviation monitoring for native-asset markets

Recommendation

Implement native-asset support by checking if the market is a native-asset market and using a dedicated address.

Alleviation

[Venus Labs, 04/09/2026]:

vBNB is not a supported market for the DeviationSentinel. The sentinel monitors ERC-20 token markets where price deviation between two oracle sources is meaningful native-asset markets like vBNB are out of scope. The `tokenConfigs` mapping is permissioned; only governance can call `setTokenConfig`. vBNB would simply never be configured, so `handleDeviation(vBNB)` would hit `revert MarketNotConfigured()` before ever reaching `underlying()`. There is no failure mode here the sentinel correctly refuses to act on unconfigured markets.

APPENDIX | VENUS LABS - EBRAKE & DEVIATIONSENTINEL CHANGES

Finding Categories

Categories	Description
Logical Issue	Logical Issue findings indicate general implementation issues related to the program logic.

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